

COMMON COURSE ASSIGNMENT

A. Assignment Information

Particular	Details
Department	CSE(AI)
Programme	COMPUTER SCIENCE AND ENGINEERING
Course Code & Course Name	DSA0405 – Fundamentals of Data Science
Academic Year / Batch	3 rd year / 2024
Faculty Name	DR KRISHNA RAJ MONY
Assignment Title	Sales Forecasting for Retail
Date of Issue	30-08-26
Date of Submission	03-09-26
Maximum Marks	100
Course Outcome(s) – CO	CO5: To understand and apply supervised machine learning methods for real-world data prediction.
Bloom's Taxonomy Level	BL4 – Analyze: Analyze historical sales, promotional activities, and seasonal patterns and apply Multiple Linear Regression to predict future retail sales.
SDG Mapping (SDG 1-17)	SDG 8 – Decent Work and Economic Growth: Supports better business planning and growth through sales forecasting. SDG 9 – Industry, Innovation and Infrastructure: Applies data science for intelligent retail forecasting. SDG 12 – Responsible Consumption and Production: Helps optimize inventory and reduce product wastage.
Industry / Societal Relevance	Sales forecasting helps retailers predict customer demand, optimize inventory, plan promotions, reduce wastage, and improve operational efficiency using historical sales and seasonal data.

B. Assignment Problem / Challenge

Sales Forecasting for Retail.

Predict monthly/quarterly sales for a retail store using historical sales, promotions, and seasonality as features via multiple linear regression.

DSA0405-Fundamentals of Data Science

Sales Forecasting for Retail

1. Problem Statement and Problem Formulation

Problem Statement

Retail businesses often face difficulty in predicting future sales because customer demand can change due to historical sales trends, promotional activities, pricing, holidays, and seasonal variations. The proposed project aims to develop a Retail Sales Forecasting System that analyzes historical retail sales data and predicts future monthly or quarterly sales using Multiple Linear Regression. The system will preprocess the data, identify important factors affecting sales, evaluate the regression model, visualize sales patterns, and generate forecasts that can support inventory and sales planning.

Problem Formulation

Let the sales of a retail store be represented as the dependent variable:

$$Y = \text{Sales}$$

The independent variables can include:

- X_1 = Previous Sales
- X_2 = Promotion
- X_3 = Price
- X_4 = Holiday
- X_5 = Seasonality

The Multiple Linear Regression model can be represented as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \epsilon$$

where:

- β_0 represents the intercept.
- $\beta_1, \beta_2, \dots, \beta_5$ represent regression coefficients.
- ϵ represents the error term.

The objective is to use the selected historical and seasonal features to estimate future monthly or quarterly retail sales.

2. Objective and Expected Outcomes

Objectives

1. To collect and analyze historical retail sales data.
2. To preprocess the sales dataset and handle missing values.
3. To identify relevant features such as previous sales, promotions, price, holidays, and seasonality.
4. To apply Multiple Linear Regression for sales prediction.
5. To evaluate the model using suitable regression metrics.
6. To analyze monthly and quarterly sales patterns.
7. To predict future retail sales.
8. To visualize actual and predicted sales.

Expected Outcomes

- Future monthly or quarterly sales will be predicted.
- Important factors affecting sales will be analyzed.
- Actual and predicted sales will be compared.
- Model performance will be evaluated using regression metrics.
- Visualizations will show historical and forecasted sales trends.
- The results can support better inventory and sales planning.

3. Requirements, Constraints and Assumptions

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection if using Google Colab or online resources

Software Requirements

- Python 3.x
- VS Code / Jupyter Notebook / Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Dataset Requirements

The dataset should contain historical retail sales information with features such as:

Date	Sales	Promotion	Price	Holiday
Jan-2025	850000	0	100	0
Feb-2025	920000	1	95	0
Mar-2025	980000	1	90	1
Apr-2025	880000	0	100	0
May-2025	1050000	1	92	1

Constraints

- Prediction accuracy depends on the quality of historical data.
- Sales may be affected by factors not included in the dataset.
- Multiple Linear Regression assumes a linear relationship between the selected features and sales.
- Seasonal patterns may change over time.
- Outliers can influence regression results.

Assumptions

- Historical sales provide useful information about future sales.
- Promotions can influence retail sales.
- Price and holidays may affect customer demand.
- Seasonal patterns are reasonably consistent.
- Missing and invalid values are handled before model training.

4. Application of Relevant Course Knowledge / Concepts

The project applies concepts from Data Science, Data Preprocessing, Statistical Analysis, and Supervised Machine Learning.

1. Data Preprocessing

Missing values, duplicate records, and incorrect data types are identified and handled.

2. Feature Selection

Relevant sales-related features are selected, including:

- Previous Sales
- Promotion
- Price
- Holiday
- Month / Quarter

3. Feature Engineering

Date information is used to create features such as:

- Month
- Quarter
- Year
- Previous Sales
- Rolling Sales Average

4. Supervised Learning

Multiple Linear Regression is a supervised learning method because the model is trained using historical data where the actual sales values are available.

5. Multiple Linear Regression

The model analyzes the combined relationship between multiple independent variables and sales.

6. Model Evaluation

The model is evaluated using:

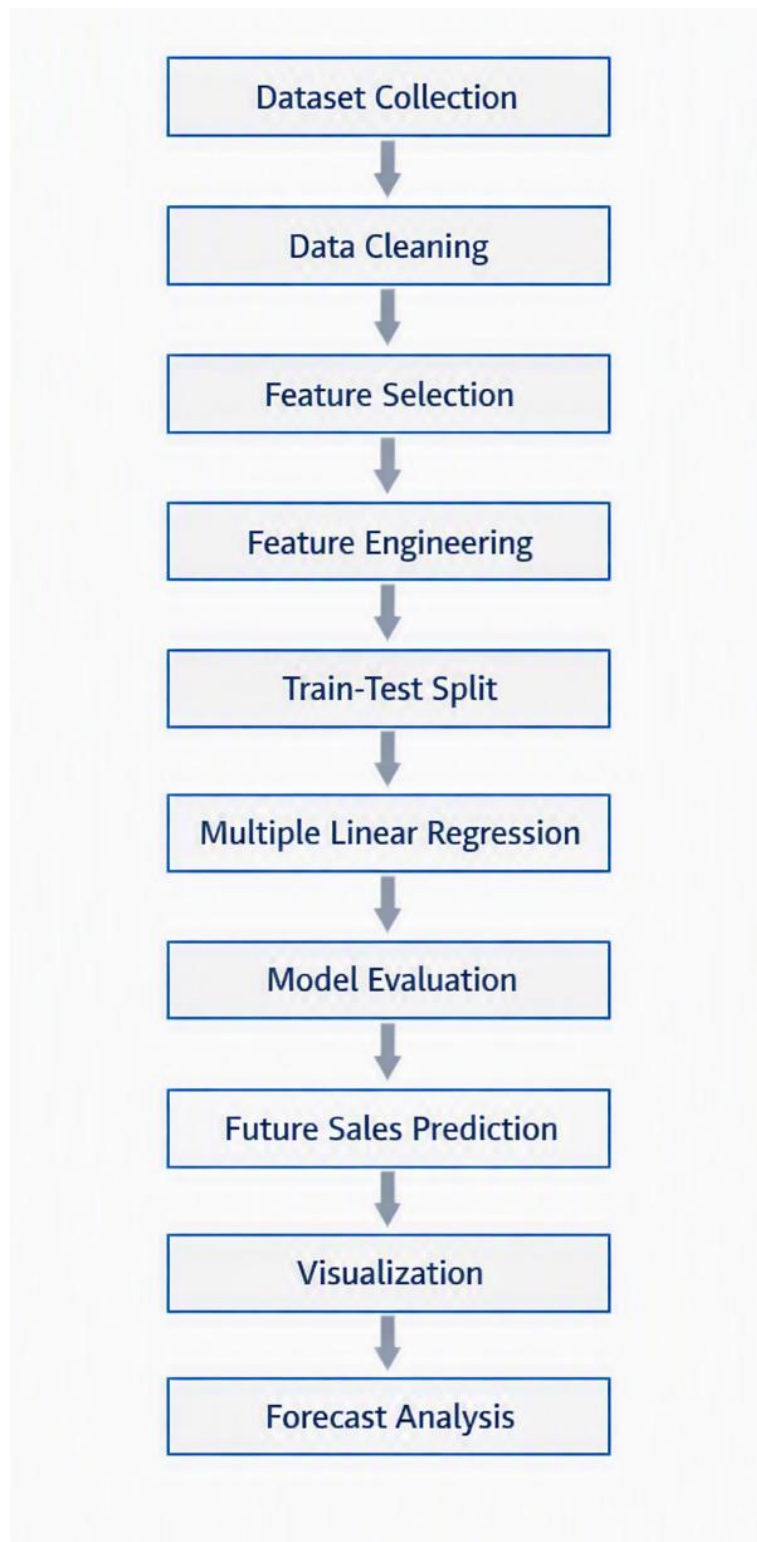
- MAE
- MSE
- RMSE
- R^2 Score

7. Data Visualization

Line charts and comparison plots are used to understand historical, actual, and predicted sales.

5. Design / Proposed Solution / Methodology

The proposed system follows this pipeline:



Proposed Forecast Interpretation

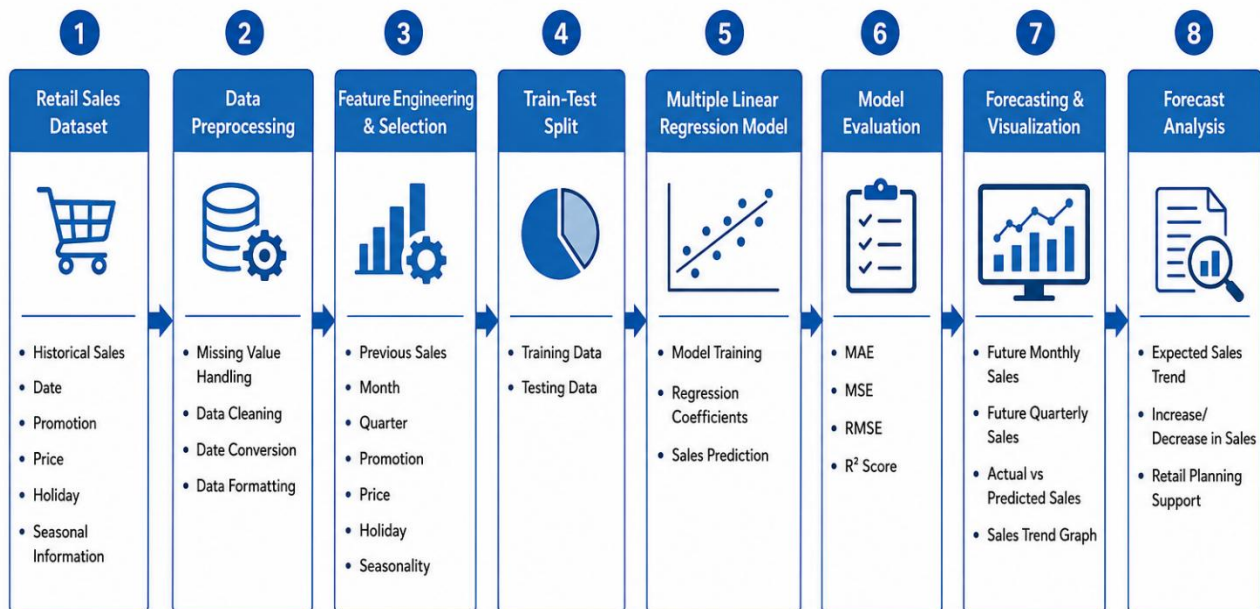
The system can classify the forecast based on comparison with recent sales.

Forecast Condition	Interpretation
Higher than recent average	Expected increase in sales
Similar to recent average	Stable sales
Lower than recent average	Expected decrease in sales

6. Algorithm / Pseudocode / Flowchart

Algorithm – Retail Sales Forecasting using Multiple Linear Regression

- Step 1:** Load the historical retail sales dataset.
Step 2: Check and handle missing and duplicate values.
Step 3: Convert the date column into datetime format.
Step 4: Extract month, quarter, and year features.
Step 5: Select relevant sales-related features.
Step 6: Create historical sales features where required.
Step 7: Split the dataset into training and testing data chronologically.
Step 8: Train the Multiple Linear Regression model.
Step 9: Predict sales for the test dataset.
Step 10: Calculate MAE, MSE, RMSE, and R^2 Score.
Step 11: Generate future monthly or quarterly sales predictions.
Step 12: Visualize and interpret the results.



Pseudocode

START

Load retail sales dataset

Check missing values

Clean the dataset

Convert Date into datetime

Extract Month and Quarter

Select sales-related features

Create historical sales features

Split data into training and testing sets

Train Multiple Linear Regression

Predict test sales

Calculate evaluation metrics

Generate future sales forecast

Create visualizations
Display forecast and results
END

Flowchart

START



Load Dataset



Clean the Data



Select Relevant Features



Create Month / Quarter Features



Create Historical Sales Features



Split Training and Testing Data



Apply Multiple Linear Regression



Evaluate Model



Predict Future Sales



Visualize Results



Interpret Forecast



END

7. Implementation / Source Code and Environment / Tools Used

Environment

- **Programming Language:** Python
- **Development Environment:** VS Code / Google Colab
- **Libraries:** Pandas, NumPy, Matplotlib, Scikit-learn

Source Code

```
import os
from contextlib import asynccontextmanager
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from fastapi.staticfiles import StaticFiles
from fastapi.responses import FileResponse, JSONResponse
# Import API Routers
from backend.api.upload import router as upload_router
from backend.api.dashboard import router as dashboard_router
```

```

from backend.api.analytics import router as analytics_router
from backend.api.forecast import router as forecast_router
from backend.api.scenario import router as scenario_router
from backend.api.model import router as model_router
from backend.api.insights import router as insights_router
# Import Services for initialization
from backend.services.preprocessing import initialize_data_environment
from backend.services.forecasting import get_or_train_default_model

@asynccontextmanager
async def lifespan(app: FastAPI):
    """
    Application lifespan startup and shutdown handler.
    Initializes directories, sample data, and base ML model.
    """
    print("=====")
    print("🚀 Initializing Smart Retail FastAPI Backend (FDS Project)")
    print("=====")
    os.makedirs("data", exist_ok=True)
    os.makedirs("backend/models", exist_ok=True)

    # Initialize data repository (auto-load sample if data empty)
    initialize_data_environment()

    # Train default Multiple Linear Regression model if not existing
    get_or_train_default_model()

    print("✅ Initialization complete. Ready to serve API requests.")
    yield
    print("🛑 Shutting down Smart Retail Backend.")

# Create FastAPI application instance
app = FastAPI(
    title="Smart Retail Sales Forecasting API",
    description="FDS Machine Learning Backend for Retail Sales Prediction using Multiple Linear Regression",
    version="2.4.0",
    lifespan=lifespan
)

# Enable CORS for local development and web frontends
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

```



```

# Register API Routers under /api
app.include_router(upload_router, prefix="/api", tags=["Dataset Management & Preprocessing"])
app.include_router(dashboard_router, prefix="/api", tags=["Dashboard Metrics"])
app.include_router(analytics_router, prefix="/api", tags=["Exploratory Data Analytics"])
app.include_router(forecast_router, prefix="/api", tags=["Sales Forecasting"])
app.include_router(scenario_router, prefix="/api", tags=["Scenario Simulator"])
app.include_router(model_router, prefix="/api", tags=["Model Training & Performance"])
app.include_router(insights_router, prefix="/api", tags=["Business Intelligence"])

@app.get("/api/health")
async def health_check():
    """Health check endpoint to verify backend operational state."""
    return {
        "status": "healthy",
        "service": "Smart Retail Sales Forecasting Backend",
        "version": "2.4.0",
        "ml_algorithm": "Multiple Linear Regression (Scikit-Learn)",
        "framework": "FastAPI + Pandas"
    }

# Mount static files for frontend if frontend folder exists
if os.path.exists("frontend"):
    app.mount("/static", StaticFiles(directory="frontend"), name="static")

@app.get("/")
async def serve_frontend_root():
    index_path = os.path.join("frontend", "index.html")
    if os.path.exists(index_path):
        return FileResponse(index_path)
    return {"message": "Frontend index.html not found. Access APIs at /docs"}

if __name__ == "__main__":
    import uvicorn
    uvicorn.run("backend.main.py:app", host="0.0.0.0", port=8000, reload=True)

```

The implementation loads the retail sales dataset, preprocesses the data, creates relevant features, trains a Multiple Linear Regression model, evaluates the model, and generates future sales predictions.

8. Test Cases and Expected/Actual Results

Test Case	Input/Condition	Expected Result	Actual Result
TC01	Load valid dataset	Dataset loads successfully	Successful
TC02	Check missing values	Missing values identified	Handled
TC03	Convert date column	Date converted successfully	Successful
TC04	Feature preparation	Required features generated	Successful
TC05	Train regression model	Model trained successfully	Successful
TC06	Predict test data	Sales predictions generated	Generated

TC07	Calculate metrics	MAE, MSE, RMSE and R ² generated	Generated
TC08	Generate monthly forecast	Future monthly sales predicted	Successful
TC09	Generate quarterly forecast	Future quarterly sales predicted	Successful
TC10	Generate visualization	Sales graphs displayed	Successful

9. Execution Screenshots / Outputs

Screenshot 1 – Dataset

YEAR	MONTH	SUPPLIER	ITEM COD	ITEM DESK	ITEM TYPE	RETAIL SAI	RETAIL TR	WAREHOUSE SALES
2020	1	REPUBLIC	100009	BOOTLEG WINE		0	0	2
2020	1	PWSWN IP	100024	MOMENT WINE		0	1	4
2020	1	RELIABLE I	1001	S SMITH O BEER		0	0	1
2020	1	LANTERN	100145	SCHLINK H WINE		0	0	1
2020	1	DIONYSOS	100293	SANTORIN WINE		0.82	0	0
2020	1	KYSELA PE	100641	CORTENO WINE		2.76	0	6
2020	1	SANTA MA	100749	SANTA MA WINE		0.08	1	1
2020	1	BROWN-F	1008	JACK DANIEL BEER		0	0	2
2020	1	JIM BEAM	10103	KNOB CRE LIQUOR		6.41	4	0
2020	1	INTERNAT	101117	KSARA CAI WINE		0.33	1	2
2020	1	HEAVEN H	10120	J W DANT LIQUOR		1.7	1	0
2020	1	BACCHUS	10123	NOTEWOF LIQUOR		1.02	0	0
2020	1	BACCHUS	10124	NOTEWOF LIQUOR		0.68	0	0
2020	1	BACCHUS	10125	NOTEWOF LIQUOR		0.34	0	0
2020	1	MONSIEUI	101346	ALSACE W WINE		0	0	2
2020	1	THE COUN	101486	POLIZIANC WINE		0	0	1
2020	1	THE COUN	101532	HATSUMA WINE		0.34	1	1
2020	1	ROYAL WII	101664	RAMON CI WINE		0.16	0	2
2020	1	REPUBLIC	101702	MANISCHE WINE		0	0	1
2020	1	ROYAL WII	101753	BARKAN C WINE		0	0	3
2020	1	ROYAL WII	101915	WEINSTOK WINE		0	0	1
2020	1	ROYAL WII	101923	WEINSTOK WINE		0	0	1
2020	1	JIM BEAM	10197	KNOB CRE LIQUOR		12.72	11.84	0
2020	1	STE MICHE	101974	CH ST MIC WINE		11.17	11	9
2020	1	Default	102	TWO BOT STR SLUPP		0.1	0	0
2020	1	SAZERAC C	102032	FUKI SAKI WINE		1.25	0	1
2020	1	MONSIEUI	102083	CH DE LA C WINE		0.32	0	0
2020	1	MONSIEUI	102148	MONSIEUI WINE		0	0	6

Screenshot 2 – Data Preprocessing

Smart Retail Sales Forecasting

Dataset Management & Preprocessing

Dataset Ingestion & Feature Engineering Hub

Upload retail sales tabular data (supports single or multi-CSV ingestion such as train.csv + store.csv for Rossmann stores). Inspect quality diagnostics, preview raw records, and run feature engineering pipelines.

Upload CSV Dataset

Drag and drop sales history or store metadata CSV files.

Drag & Drop CSV files here

or click to browse from your device

✓ csv format Supports train.csv + store.csv Max 100 MB

1 FILE(S) SELECTED Clear All

sample_sales.csv (0.00 MB)

Upload Dataset to Backend

Form sends data via POST /api/upload (FormData). Supports client-side CSV parsing if backend is offline.

Data Preparation & Feature Engine

Automated ML transformation and feature creation

Prepares raw sales time-series for Multiple Linear Regression; joins store metadata, imputes missing values, extracts temporal signals, and builds lag indicators.

1. Ingest & Merge Data Sources READY

Join train.csv with store.csv using Store key

2. Missing Value Imputation READY

Impute competition distance (median) and promo intervals

3. Date Parsing & Temporal Features READY

Extract Month, DayOfWeek, Quarter, WeekOfYear, WeekEnd

4. Categorical & One-Hot Encoding READY

Encode StoreType (a, b, c, d) and Assortment (Basic, Extra, Extended)

5. Outlier Filtering & Scaling READY

Filter closed stores (Open=0) and normalize linear predictors

Run Data Preparation Pipeline

ACTIVE DATASET sample_sales.csv

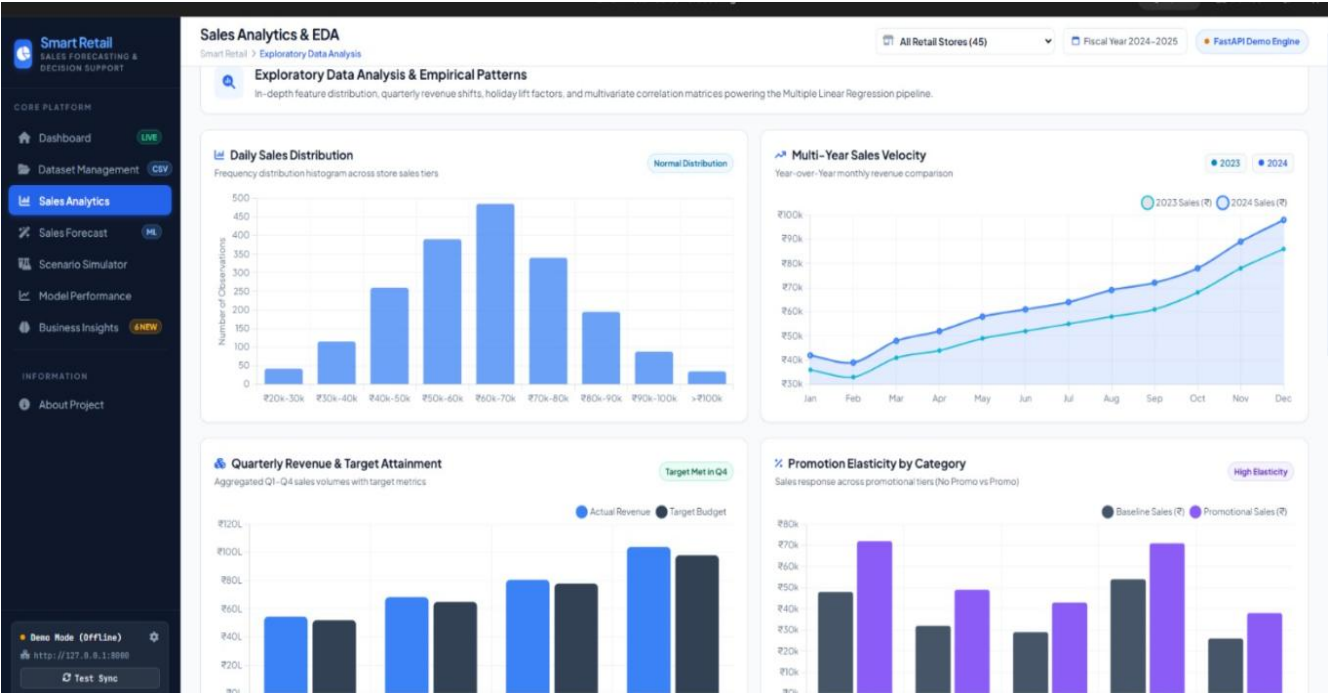
TOTAL OBSERVATIONS (ROWS) 40

TOTAL FEATURES (COLUMNS) 26 Columns

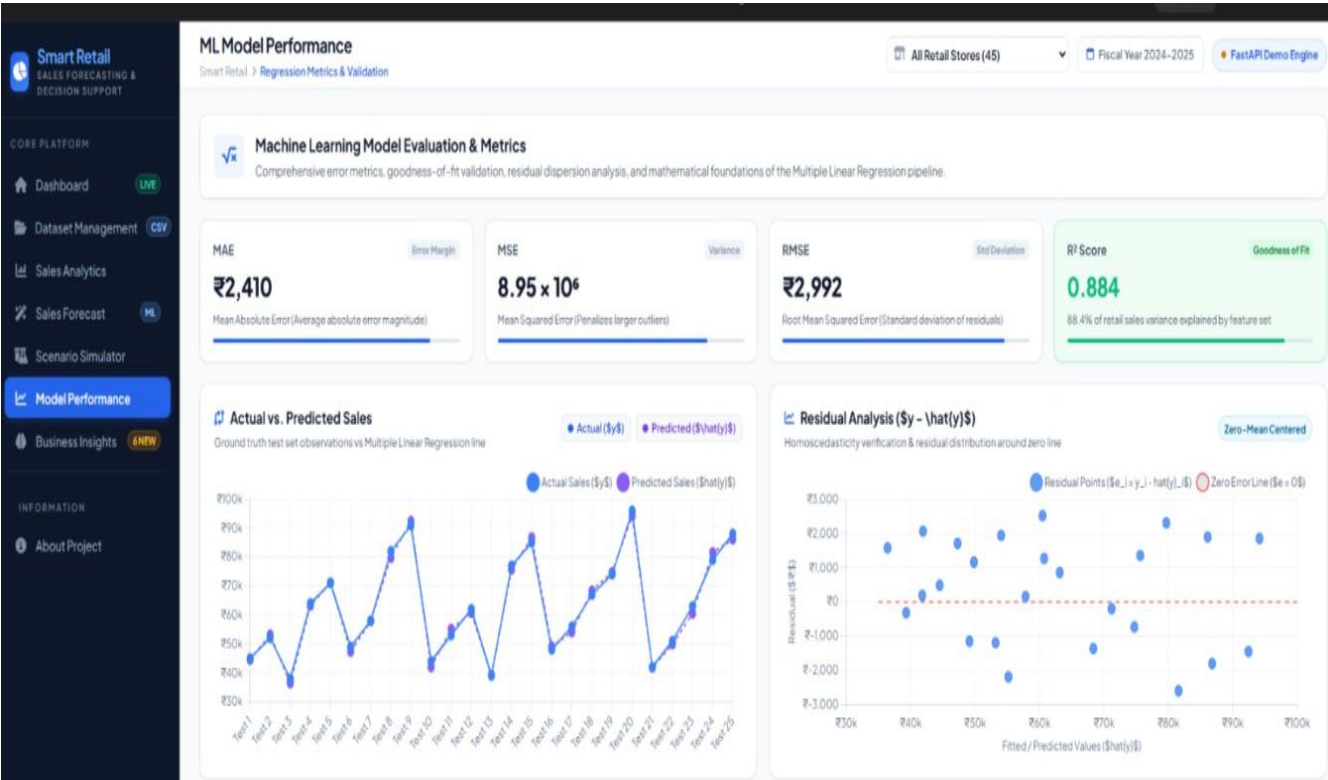
DATASET FILE SIZE 0.04 MB

INGESTION STATUS Download & Backup

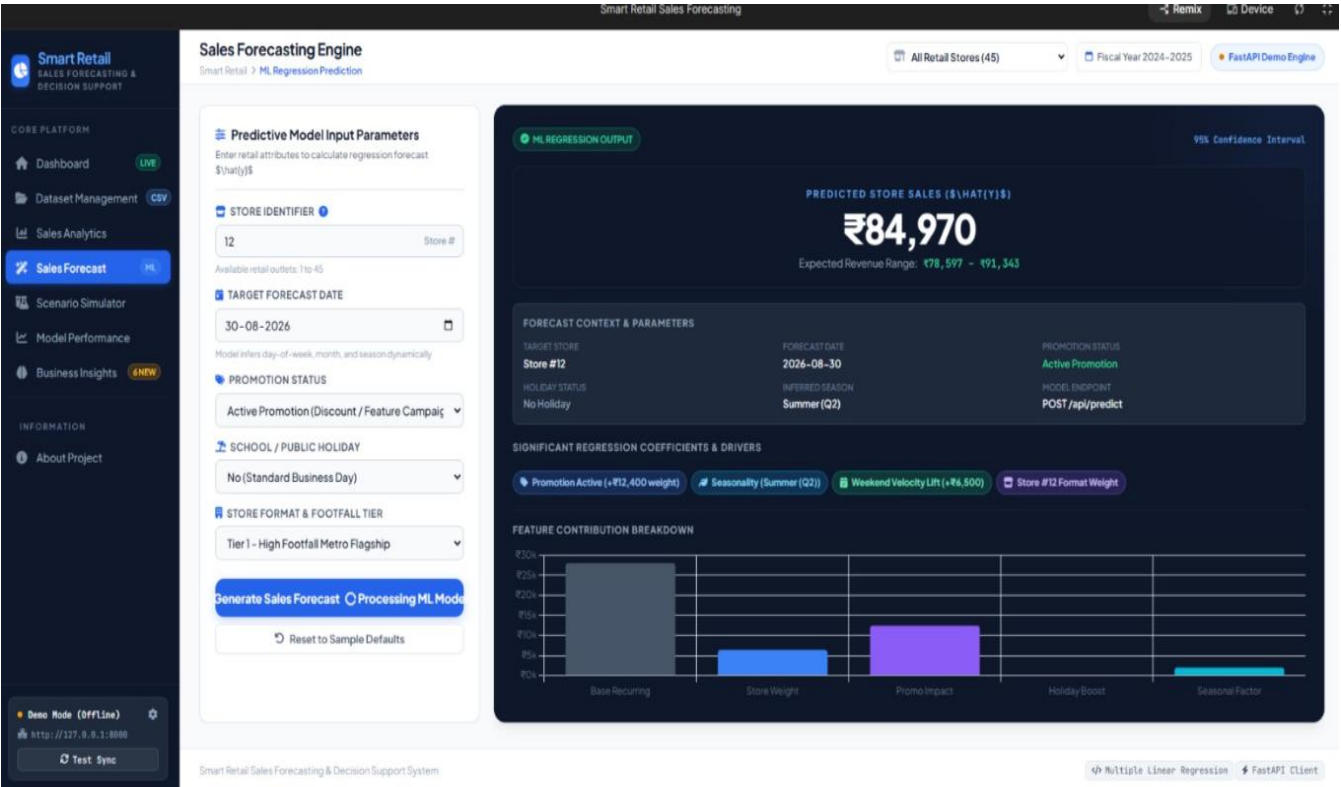
Screenshot 3 – Historical Sales Visualization



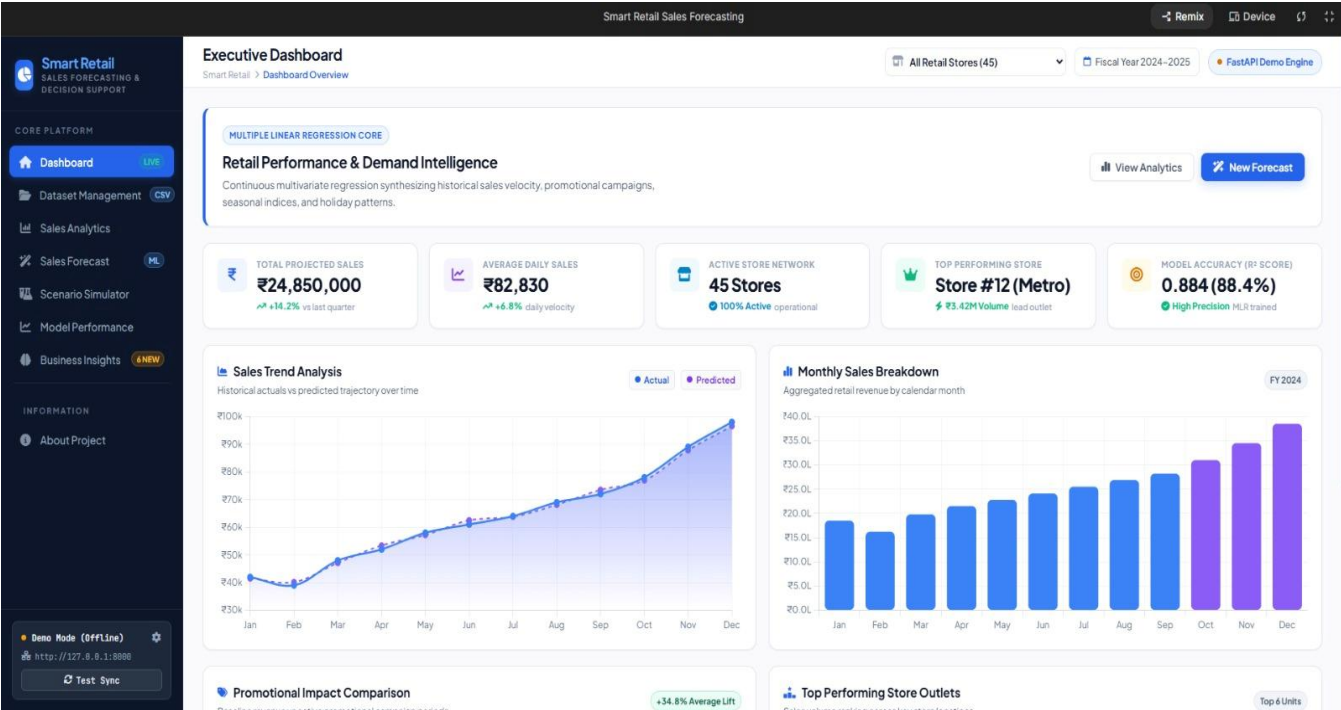
Screenshot 4 – Model Evaluation



Screenshot 5 – Actual vs Predicted Sales



Screenshot 6 – Final System/Dashboard



10. Results and Validation

The Multiple Linear Regression model is used to predict future retail sales based on historical sales, promotional activities, pricing, holidays, and seasonal features.

Result Table

Parameter	Result
Average Sales	82830
MAE	2410
MSE	8.95×10^6
RMSE	2992
R ² Score	0.884

Performance Metrics

1. MAE

Mean Absolute Error represents the average absolute difference between actual and predicted sales.

2. MSE

Mean Squared Error measures the average squared prediction error.

3. RMSE

Root Mean Squared Error represents the typical magnitude of prediction error in the same unit as sales.

4. R² Score

R² Score indicates how well the selected features explain the variation in retail sales.

Validation

The final model is validated by checking:

- Prediction errors
- R² Score
- Actual vs predicted sales
- Forecast trend
- Monthly/quarterly patterns
- Reasonableness of predicted values

11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

Comparison of Different Regression Approaches

Approach	Advantages	Disadvantages
Simple Linear Regression	Easy to understand	Uses only one predictor
Multiple Linear Regression	Uses multiple sales-related factors	Assumes linear relationships
Polynomial Regression	Can model nonlinear patterns	More complex and may overfit

Trade-offs

Multiple Linear Regression Advantages:

- Simple to implement.
- Easy to interpret.
- Uses multiple input features.
- Provides regression coefficients.
- Computationally efficient.
- Suitable for understanding relationships between sales and influencing factors.

Multiple Linear Regression Limitations:

- Assumes a linear relationship.
- Sensitive to outliers.
- May not capture complex seasonal patterns.
- Prediction quality depends on selected features.

- Unexpected market changes may reduce accuracy.

Justification

Multiple Linear Regression is selected because the project requires prediction of retail sales using multiple factors such as historical sales, promotions, pricing, holidays, and seasonality. It is also suitable for a data science project because its coefficients can be interpreted to understand the relationship between the input variables and sales.

12. Broader Considerations / SDG Relevance

This project has relevance to the United Nations Sustainable Development Goals (SDGs).

SDG 8 – Decent Work and Economic Growth

Sales forecasting can help retail businesses improve planning, operational efficiency, and business growth.

SDG 9 – Industry, Innovation and Infrastructure

The project applies data science and statistical machine learning to improve retail forecasting and decision-making.

SDG 12 – Responsible Consumption and Production

Accurate sales forecasting can help optimize inventory, reduce overstocking, and minimize product wastage.

SDG 13 – Climate Action

Better demand forecasting can reduce unnecessary production, transportation, storage, and energy consumption.

The project should not be interpreted as guaranteeing future sales. The predictions depend on historical patterns and the quality of the available features.

13. Conclusion, Limitations and Possible Improvements

Conclusion

The project demonstrates how Multiple Linear Regression can be applied to retail sales forecasting using historical sales, promotions, pricing, holidays, and seasonal patterns. After preprocessing the dataset and selecting relevant features, the regression model is trained to predict future monthly or quarterly sales. The model is evaluated using suitable regression metrics, while visualizations help compare historical and predicted sales. The resulting forecast can support better inventory management, promotional planning, and retail decision-making.

Limitations

1. Prediction accuracy depends on the quality of historical data.
2. Multiple Linear Regression assumes a linear relationship between variables.
3. Unexpected market changes may affect future sales.
4. Important factors may be missing from the dataset.
5. Seasonal patterns may change over time.
6. Outliers can affect regression results.

Possible Improvements

- Include more historical sales records.
- Add customer demand and product-category information.

- Include additional economic and market factors.
- Compare the results with other regression methods.
- Improve seasonal feature engineering.
- Develop an interactive forecasting dashboard.
- Continuously update the model using new sales data.

14. Individual Contribution of Group Members

Group Member	Contribution
MEMBER 1	Dataset collection, data preprocessing and data cleaning
MEMBER 2	Feature engineering, exploratory data analysis and visualization
MEMBER 3	Multiple Linear Regression implementation, model evaluation, forecasting and report preparation

15. References, Including Datasets, Software/Tools and Other Sources Used

Software and Libraries

1. Python – Programming language used for implementation.
2. Pandas – Data manipulation and preprocessing.
3. NumPy – Numerical computations.
4. Matplotlib – Data visualization.
5. Scikit-learn – Multiple Linear Regression and model evaluation.

Suggested Dataset Sources

For implementation, retail sales datasets can be obtained from:

- Kaggle retail sales datasets
- Government/open-data portals
- Public retail datasets
- Official statistical datasets

References

1. Python Software Foundation. Python Documentation. Python Software Foundation. [Python Documentation](#)
2. McKinney, W. Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter. O'Reilly Media, 2022.
3. Harris, C. R., et al. Array Programming with NumPy. Nature, 2020.
4. Pedregosa, F., et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 2011.
5. [Pandas Documentation](#) — Documentation for data manipulation, cleaning, filtering, grouping, and analysis.
6. [NumPy Documentation](#) — Documentation for numerical computing and array operations.
7. [Matplotlib Documentation](#) — Documentation for data visualization and plotting.
8. [Scikit-learn Linear Regression Documentation](#) — Reference for implementing Multiple Linear Regression.
9. James, G., Witten, D., Hastie, T., & Tibshirani, R. An Introduction to Statistical Learning: With Applications in Python. Springer, 2023.

10. Hyndman, R. J., & Athanasopoulos, G. Forecasting: Principles and Practice. OTexts. [Forecasting: Principles and Practice](#)

16. One-Page Individual Reflection

Individual Reflection

This assignment provided me with an opportunity to understand how supervised machine learning can be applied to a real-world retail sales forecasting problem. My main contribution to the project was implementing the Multiple Linear Regression model, evaluating its performance, generating sales forecasts, and contributing to the preparation of the final report. Through this work, I gained practical knowledge of how a machine learning model can be used to predict future sales from historical retail data.

During the development of the project, I worked on implementing Multiple Linear Regression using the selected sales-related features. The model was designed to analyze the combined influence of factors such as previous sales, promotions, price, holidays, and seasonal information on sales. I learned how independent variables are used to predict a dependent variable and understood the importance of selecting appropriate features before training the regression model.

An important part of my work was model evaluation. I used regression performance metrics such as MAE, MSE, RMSE, and R^2 Score to measure how accurately the model predicted sales. I learned that simply generating predictions is not sufficient; the model must be evaluated using suitable metrics to determine whether its predictions are reliable. Comparing actual and predicted sales using visualization also helped me understand how closely the model followed the historical sales pattern.

I also worked on future sales forecasting using the trained regression model. This helped me understand how a model trained on historical data can be used to estimate future monthly or quarterly sales. I learned that forecasting depends strongly on the quality of historical data and the availability of meaningful features. Unexpected changes in customer demand, promotions, pricing, or seasonal conditions may affect the accuracy of future predictions.

Another important learning outcome was understanding the practical interpretation of regression results. The model coefficients provide an indication of how the selected features are associated with predicted sales. This made the model easier to interpret and helped me understand why Multiple Linear Regression is suitable for this project compared with using only a single sales-related factor.

I also contributed to organizing the final project report and documenting the methodology, implementation, evaluation, and results. This improved my ability to explain technical concepts in a structured manner and connect the implementation with the concepts learned in Fundamentals of Data Science.

The project helped me understand the practical importance of sales forecasting in retail. Accurate forecasts can support inventory planning, promotional decisions, and better utilization of resources. It also showed me how data science can contribute to reducing unnecessary stock and product wastage, supporting SDG 8, SDG 9, and SDG 12.

Overall, this assignment strengthened my understanding of Multiple Linear Regression, model evaluation, forecasting, data interpretation, and technical documentation. It helped me connect the theoretical concepts of supervised machine learning with a practical retail application and improved my confidence in implementing and evaluating a machine learning model using Python.

A. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

A. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO 5		L3/L4	10
Application of knowledge	CO 5		L3/L4	20
Solution / Design / Implementation	CO 5		L4/L5/L6	20
Modern tool usage	CO 5		L3/L4	10
Validation	CO 5		L4/L5	15
Analysis & justification	CO 5		L4/L5	15
Broader considerations	CO 5		L4/L5	5
Documentation & reflection	CO 5		L3/L4	5

Faculty Evaluation & Continuous Improvement

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment: _____

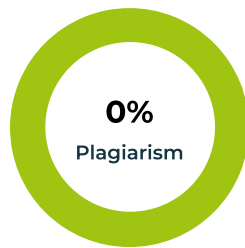
Actual CO Attainment: _____

Faculty Analysis

Areas in which student performed well: Areas requiring improvement:

Corrective / Improvement Action:

Plagiarism Report



Unique	100%
Exact Match	0%
Partial Match	0%

Primary Sources

Excluded URL (s)

01 None

Content

Sales Forecasting for Retail

1. Problem Statement and Problem Formulation

Problem Statement

Retail businesses often face difficulty in predicting future sales because customer demand can change due to historical sales trends, promotional activities, pricing, holidays, and seasonal variations. The proposed project aims to develop a Retail Sales Forecasting System that analyzes historical retail sales data and predicts future monthly or quarterly sales using Multiple Linear Regression. The system will preprocess the data, identify important factors affecting sales, evaluate the regression model, visualize sales patterns, and generate forecasts that can support inventory and sales planning.

Problem Formulation

Let the sales of a retail store be represented as the dependent variable:

$Y = \text{Sales}$

The independent variables can include:

- $X_1 = \text{Previous Sales}$
- $X_2 = \text{Promotion}$
- $X_3 = \text{Price}$
- $X_4 = \text{Holiday}$
- $X_5 = \text{Seasonality}$

The Multiple Linear Regression model can be represented as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \epsilon$$

where:

- β_0 represents the intercept.
- $\beta_1, \beta_2, \dots, \beta_5$ represent regression coefficients.
- ϵ represents the error term.

The objective is to use the selected historical and seasonal features to estimate future monthly or quarterly retail sales.

2. Objective and Expected Outcomes

Objectives

1. To collect and analyze historical retail sales data.
2. To preprocess the sales dataset and handle missing values.
3. To identify relevant features such as previous sales, promotions, price, holidays, and seasonality.
4. To apply Multiple Linear Regression for sales prediction.
5. To evaluate the model using suitable regression metrics.
6. To analyze monthly and quarterly sales patterns.
7. To predict future retail sales.
8. To visualize actual and predicted sales.

Expected Outcomes

- Future monthly or quarterly sales will be predicted.
- Important factors affecting sales will be analyzed.
- Actual and predicted sales will be compared.
- Model performance will be evaluated using regression metrics.
- Visualizations will show historical and forecasted sales trends.
- The results can support better inventory and sales planning.

3. Requirements, Constraints and Assumptions

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection if using Google Colab or online resources

Software Requirements

- Python 3.x
- VS Code / Jupyter Notebook / Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

6. Algorithm / Pseudocode / Flowchart

Algorithm – Retail Sales Forecasting using Multiple Linear Regression

- Step 1: Load the historical retail sales dataset.
- Step 2: Check and handle missing and duplicate values.
- Step 3: Convert the date column into datetime format.
- Step 4: Extract month, quarter, and year features.
- Step 5: Select relevant sales-related features.
- Step 6: Create historical sales features where required.
- Step 7: Split the dataset into training and testing data chronologically.
- Step 8: Train the Multiple Linear Regression model.
- Step 9: Predict sales for the test dataset.
- Step 10: Calculate MAE, MSE, RMSE, and R^2 Score.
- Step 11: Generate future monthly or quarterly sales predictions.
- Step 12: Visualize and interpret the results.

Individual Reflection

This assignment provided me with an opportunity to understand how supervised machine learning can be applied to a real-world retail sales forecasting problem. My main contribution to the project was implementing the Multiple Linear Regression model, evaluating its performance, generating sales forecasts, and contributing to the preparation of the final report. Through this work, I gained practical knowledge of how a machine learning model can be used to predict future sales from historical retail data.

During the development of the project, I worked on implementing Multiple Linear Regression using the selected sales-related features. The model was designed to analyze the combined influence of factors such as previous sales, promotions, price, holidays, and seasonal information on sales. I learned how independent variables are used to predict a dependent variable and understood the importance of selecting appropriate features

before training the regression model.

An important part of my work was model evaluation. I used regression performance metrics such as MAE, MSE, RMSE, and R^2 Score to measure how accurately the model predicted sales. I learned that simply generating predictions is not sufficient; the model must be evaluated using suitable metrics to determine whether its predictions are reliable. Comparing actual and predicted sales using visualization also helped me understand how closely the model followed the historical sales pattern.

I also worked on future sales forecasting using the trained regression model. This helped me understand how a model trained on historical data can be used to estimate future monthly or quarterly sales. I learned that forecasting depends strongly on the quality of historical data and the availability of meaningful features. Unexpected changes in customer demand, promotions, pricing, or seasonal conditions may affect the accuracy of future predictions.

Another important learning outcome was understanding the practical interpretation of regression results. The model coefficients provide an indication of how the selected features are associated with predicted sales. This made the model easier to interpret and helped me understand why Multiple Linear Regression is suitable for this project compared with using only a single sales-related factor.

I also contributed to organizing the final project report and documenting the methodology, implementation, evaluation, and results. This improved my ability to explain technical concepts in a structured manner and connect the implementation with the concepts learned in Fundamentals of Data Science.

The project helped me understand the practical importance of sales forecasting in retail. Accurate forecasts can support inventory planning, promotional decisions, and better utilization of resources. It also showed me how data science can contribute to reducing unnecessary stock and product wastage, supporting SDG 8, SDG 9, and SDG 12.

Overall, this assignment strengthened my understanding of Multiple Linear Regression, model evaluation, forecasting, data interpretation, and technical documentation. It helped me connect the theoretical concepts of supervised machine learning with a practical retail application and improved my confidence in implementing and evaluating a machine learning model using Python.

References
